Lab 3
Component Modelling & Architectural Pattern Selection

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Scenario: 26

System:
Peer-to-Peer Carpooling & Expense Splitter

Architecture Selection:
We chose Layered Architecture for the Peer-to-Peer Carpooling & Expense Splitter System.

Architectural Choice

The system follows a Layered Architecture consisting of Presentation, Business, and Data layers. The Presentation Layer provides the Carpool interface for both Rider Commuters and Driver Hosts. The Business Layer includes the Ride Manager, Route Matching Service, and Payment Service, which handle the main system operations. The Data Layer consists of the Carpool Database, which stores information such as users, rides, routes, and travel history.

Reason 1

A Layered Architecture creates a clear separation between different responsibilities within the system. The user interface is kept separate from business processes such as managing rides, checking route compatibility, and calculating shared expenses. This separation makes the system easier to understand, develop, maintain, and update.

Reason 2

This architecture is appropriate because the system's major functions can be divided into specialised components that communicate through well-defined interfaces. For example, the Route Matching Service can determine whether routes overlap while considering the 1.5 km detour tolerance, whereas the Payment Service can manage the automatic calculation and sharing of fuel costs.

Security Advantage

The layered structure also improves security by keeping sensitive information within the Data Layer and preventing direct access from the user interface. Access can instead be controlled through the appropriate business components. This is particularly important for protecting information such as user ratings, driving license verification documents, and travel histories that the system needs to store securely.

Performance Benefit

Separating the system into layers allows each component to focus on its specific responsibilities. For instance, route compatibility calculations can be processed by the Route Matching Service without placing additional workload on the Carpool UI. This reduces unnecessary processing in the presentation layer and helps provide a more responsive experience for users.